COMPUTER SCIENCE

O Level Notes

Syllabus 2210 / 0478

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O Level Computer Science

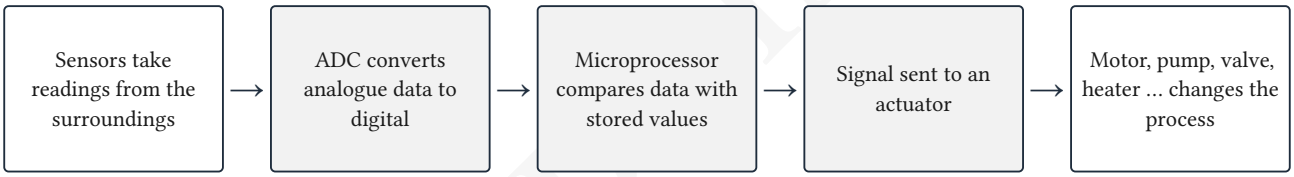
Chapter 6: Automated and Emerging Technologies

6.1 Automated Systems · 6.2 Robotics · 6.3 Artificial Intelligence

6.1.1 Sensors, Microprocessors and Actuators

An **automated system** is a combination of **software and hardware** (sensors, microprocessors and actuators) that is designed and programmed to **work automatically without human intervention**. A human often still **monitors** the system.

- **Sensors** are **input devices** that take readings from the surroundings and send the data to a microprocessor or computer. If the data is **analogue**, an **ADC (analogue to digital converter)** changes it to digital first.
- The **microprocessor** processes the data, compares it with stored values and decides what action to take, based on its program.
- **Actuators** are **output devices**. They receive signals from the microprocessor and control motors, pumps, valves, wheels, solenoids, heaters and so on.



The sensor, microprocessor and actuator cycle (it repeats continuously)

> **Exam Tip:** For every automated system, describe the **same cycle**: sensors read the data → ADC converts it → microprocessor compares with stored values → signal to actuator → action. Then the sensors read again, so it is **continuous**. Always **name the sensor** and **name what the actuator does**.

6.1.2 Advantages and Disadvantages of Automated Systems

These points apply to **almost every** automated system.

Advantages	Disadvantages
• **Faster** than a human at taking action • **Safer**: humans are kept away from dangerous places and the system reacts in time • Runs at **optimum conditions** because small changes are found and corrected quickly • **Cheaper in the long run**: fewer workers are needed and it works **24/7** • **More consistent**, higher productivity and less wasted material • Can be **monitored remotely** by one supervisor	• **Expensive to set up** and needs a lot of testing • Unexpected conditions that were **never tested** may cause safety problems (so a supervisor is needed) • Can be attacked by **hackers** (cyberattacks) • Needs **more maintenance**, often by specialist technicians • **Job losses** and loss of skills • Faulty or dirty sensors give **false data**

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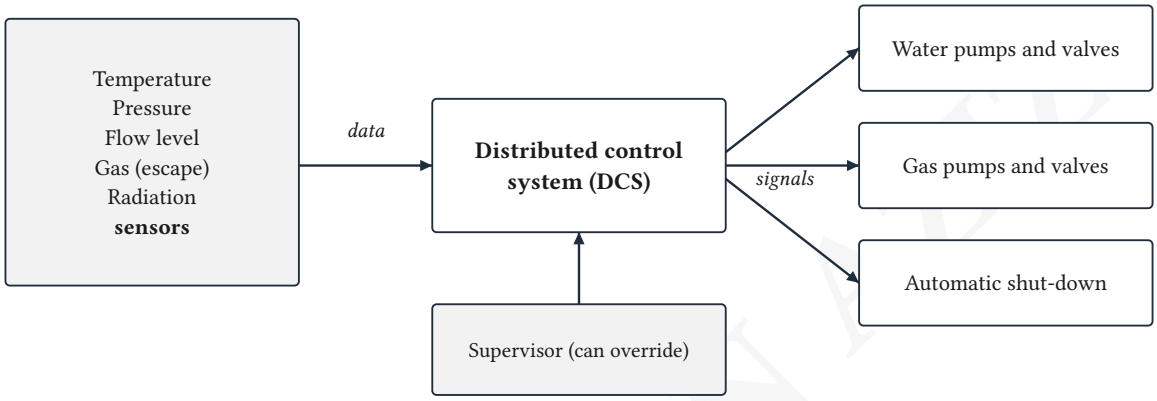
> **Exam Tip:** In exam answers, **relate the point to the scenario**. Do not just write "it is faster". Write "it reacts faster than a human operator if the pressure becomes too high".

Industrial Applications

Automation has improved quality. For example, in car engine manufacture, a manual piston fitting error rate of about 1.5% fell to 0.00001% when automated.

Example 1: Nuclear power station

A **distributed control system (DCS)** is a powerful computer that monitors and controls the whole process with **no human interaction** needed.



Nuclear power station: sensors, DCS and actuators

- **Sensors:** temperature, pressure, flow level, gas (escape) and radiation level.
- Data goes to the DCS (through an ADC if analogue). The DCS checks it against a **database of operating parameters**.
- If action is needed, signals go to **actuators** that operate **water pumps, gas pumps, valves** or an **automatic shut-down** system.
- A **supervisor** in a remote control room watches a screen and can **override** the DCS and shut the process down.

Advantages	Disadvantages
• Much faster than a human at taking action • Safer: keeps workers away from radiation • Process runs under optimum conditions • Cheaper in the long run (no large team watching 24 hours a day)	• Very expensive to set up and test • A situation not considered in testing could be unsafe • Can be hacked (best protected by having no external links to the DCS) • Needs expensive extra maintenance

Example 2: Manufacture of paracetamol

Process 1 makes the paracetamol and **process 2** makes the solid tablets. Sensors (for example temperature, pressure and level sensors) send data to a **central computer**, which checks a **database** to make sure both processes are inside the correct limits. The computer sends signals to **actuators** that operate **pumps, valves, heaters, stirrers and pistons**. A remote **monitoring station** has an operator who can override the computer.

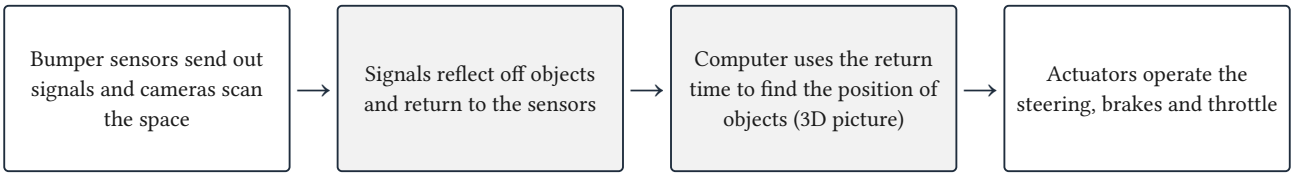
Extra advantages: more efficient use of materials, higher productivity, more consistent tablets. The disadvantages are the same as the nuclear example.

Transport

Automatic train signal control: sensors beside the track send signals to the on-board computer. If the driver passes a **red signal**, the computer makes **actuators apply the brakes**.

Autopilot in aircraft: sensors and actuators control the **wing flaps, throttle and rudder** to keep the correct **height, speed and direction**.

Example 3: Self-parking cars



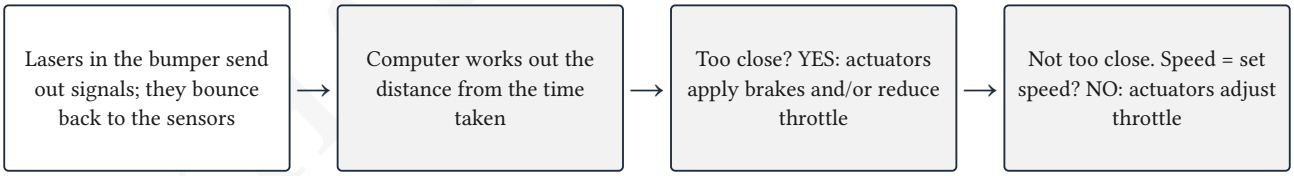
A self-parking car

- The driver drives along the row of parked cars. **Sensors and cameras** measure the size of spaces and the computer warns the driver when a suitable one is found.
- The driver selects auto-park. The computer then controls the **steering, brakes and throttle** through actuators.
- Bumper sensors are both **transmitters and receivers**. The **time taken for the signal to return** gives the position of objects, which gives the computer a 3D image of the surroundings.
- If a **new object** appears (for example a child), the manoeuvre stops.

Advantages	Disadvantages
• Cars fit into **smaller spaces**, so fewer parking spaces are needed • Less traffic disruption (a manual car takes several seconds to park) • Fewer dents and scratches • Safer: sensors monitor all objects • Consistent results	• Driver **over-relies** on the system (loss of skills) • Dirty or faulty sensors and cameras send wrong data • Sensors may **miss low kerbs** • Expensive option and needs extra maintenance

Example 4: Adaptive cruise control

The driver sets a **cruising speed**. The car then keeps a **safe distance** from the vehicle in front.



Adaptive cruise control (the process repeats all the time)

- The computer compares the **distance** with the **safe stopping distance** for the current speed. A higher speed needs a bigger gap.

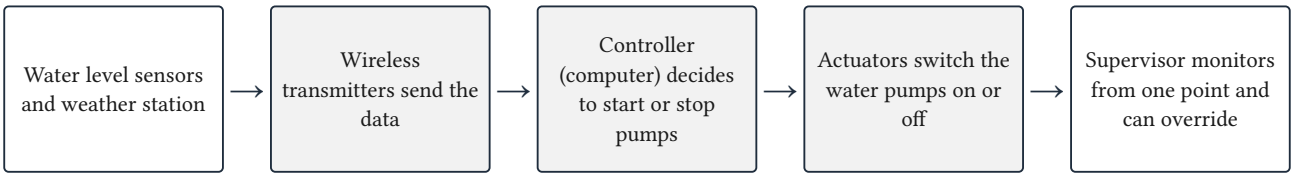
Speed (kph)	40	50	60	70	80	90	100	110	120
Stopping distance (m)	17	24	32	42	53	64	77	92	107

> **Exam Tip:** If the question gives a table of stopping distances, say: the computer **looks up the stopping distance for the current speed**, **compares it with the measured distance**, and **signals the actuators to brake** if the distance is smaller.

Agriculture

Example 5: Automatic irrigation (Brazil)

Irrigation is fully automatic and uses **wireless transmission**, so it works in very large, remote fields (some are over 10 km²).



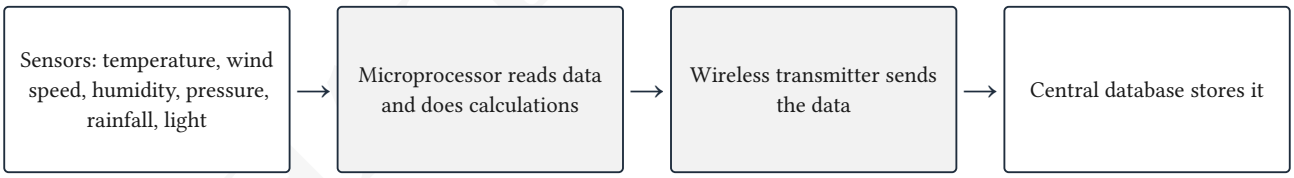
Automatic irrigation system

- An **automatic weather station** sends data to the controller **every ten minutes**. This matters when very wet or very dry weather is detected or predicted.
- **Ultrasonic water level sensors** measure the water in the irrigation channels.
- The controller uses both sets of data to decide whether to **start or stop pumps** through actuators.
- Different crops need different conditions (rice needs flooding, orange trees need dry conditions), so each controller can be programmed differently.

Advantages	Disadvantages
• Lower **labour costs**: one supervisor watches vast areas • Better, more efficient use of **water** • Faster than a person checking many kilometres of channels • Safer (fields may be 40 °C) • Different crops can be given different conditions at the same time	• Expensive to set up • High maintenance cost and **specialist technicians** may be far away • A **blocked or collapsed channel** is not detected, so some areas get too much water and some too little

Weather Stations

Automated weather stations save labour and collect data from **remote areas**. They need a **microprocessor, storage (database), a battery (often solar charged)** and sensors.



Automated weather station (powered by a battery with solar charging)

Sensor / device	Measures
Thermometer (temperature sensor)	Temperature
Anemometer	Wind speed
Hygrometer (humidity sensor)	Humidity
Barometer (pressure sensor)	Air pressure
Level sensor	Rainfall
Light sensor	Hours of daylight

The microprocessor does calculations (for example hours of daylight, actual rainfall, wind direction) and stores the results in a central database. Stations near airports send reports to pilots **every five minutes**.

The **only actuator** is in the **tipping bucket rain gauge**. At set time intervals the microprocessor tells an actuator to operate a piston that tips the bucket. The water goes into a vessel where **level sensors** measure how much rain fell.

Gaming

Sensors give games more **realism**:

- **Accelerometers** measure **acceleration and deceleration** and so detect **tilting** the device forward, backward and side to side.
- **Proximity sensors** (in touch pads) detect the **position of a hand or finger**.

A microcontroller uses this data so the player's real movements are copied in the game, which gives a more **immersive** experience.

Lighting

Example 6: Garden lighting at a house

- A **light sensor** detects when it becomes dark, so the microprocessor sends signals (through an interface) to **switch on LED lights** around the garden.
- An **infrared sensor** detects movement and is used as a **security device** when the house is empty.
- When it is dark, a **fountain light show** can start. The microprocessor sends signals to **actuators** that turn the water pumps on and off.

Advantages	Disadvantages
• Lights are controlled automatically • **Lower energy use** (lights are on only when needed) • **Longer bulb life** • Wireless connection is safer (no trailing wires) • New light displays can be programmed	• Expensive to set up • Wireless connections are **less reliable** than wired • Needs more maintenance

Science

Example 7: Chemical process in a laboratory

A burette ('A') adds one chemical to a reaction vessel ('B'). When the reaction is finished the mixture turns **bright orange**.

- **Level sensors** measure how much liquid has been added from 'A'.
- A **colorimeter** checks the **colour** of the chemical in 'B'.
- The microprocessor receives both readings and sends signals to an **actuator** that **opens or closes the tap** on 'A'.

Advantages	Disadvantages
• **Repeatable** (consistent) results • Less dangerous if the chemicals are hazardous • Faster: several experiments can run at once • Automatic analysis of results • Fewer highly trained staff needed • Can be monitored from anywhere in the world	• Less flexible than human technicians • Security risk if data is shared globally • Equipment is expensive to buy and set up

Summary of the Examples

Example	Sensors used	Actuators operate	Computer / microprocessor

Nuclear power station	Temperature, pressure, flow level, gas, radiation	Water and gas pumps, valves, shut-down	DCS compares with database and decides the action
Paracetamol manufacture	Temperature, pressure, level	Pumps, valves, heaters, stirrers, pistons	Checks both processes against the database
Self-parking car	Bumper (transmit and receive) sensors, cameras	Steering, brakes, throttle	Calculates object positions from signal return time
Adaptive cruise control	Laser distance sensors	Brakes, throttle	Calculates distance, compares with safe distance and set speed
Irrigation	Ultrasonic water level sensors, weather station	Water pumps	Controller starts or stops pumps
Weather station	Temperature, wind, humidity, pressure, level, light	Piston for the tipping bucket	Calculates values and stores them
Garden lighting	Light sensor, infrared sensor	Fountain pumps (lights through an interface)	Switches lights and fountain show on and off
Laboratory experiment	Level sensors, colorimeter	Tap on the burette	Controls the tap using both readings
Train doors	Proximity sensors, cameras	Motors that open and close the doors	Opens, sounds alarm after 30 s, re-opens if someone is in the way

AI in automated systems: many modern systems also use AI. It can **store and learn from huge amounts of data** and **spot patterns humans miss**, but it **depends on the quality of the data that trained it**.

6.2.1 What is Robotics?

Robotics is the branch of (computer) science that brings together the **design, construction and operation of robots**. A **robot** is a mechanical device that can carry out tasks normally done by humans.

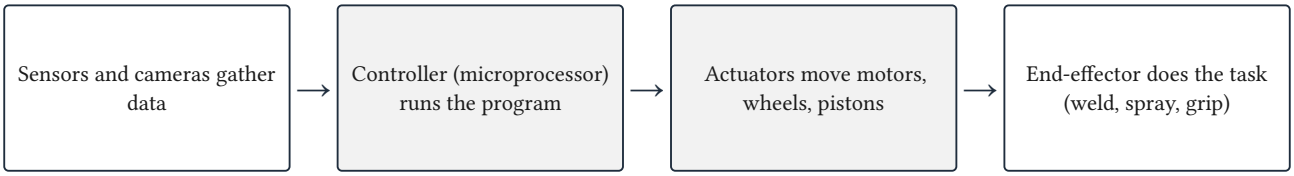
Where robots are found:

- **Factories:** welding, spray painting car panels, fitting windscreens, cutting metal parts, bottling and labelling, warehouses.
- **Home:** autonomous floor sweepers, lawn mowers, pool cleaners, window cleaners.
- **Drones (UAVs):** remote controlled or autonomous. Used for aerial photographs (reconnaissance) and parcel delivery.

6.2.2 Characteristics of a Robot

To be called a robot a machine must have **all three**:

Characteristic	Details
1. Senses its surroundings	Uses **sensors** (light, pressure, temperature, acoustic …) to find size, shape, weight, hot or cold. All data goes to the microprocessor
2. Has a degree of movement	Uses wheels, cogs, pistons, gears, motors and actuators to turn, twist, move, grip or lift. May have **end-effectors**
3. Is programmable	Has a **controller** (its brain) that decides the action. The controller is programmable so the robot can do different tasks



The three characteristics of a robot working together

An **end-effector** is an **attachment on a robot arm** that lets it do a specific task, for example a welding torch, a spray gun, a cutter or a gripper.

> **Important notes**
> - Many robots are **not** artificially intelligent. They do **repetitive tasks** and do not need adaptive, human-like characteristics. Robotics and AI are two different fields.
> - **Software robots** such as **search engine bots (web crawlers)** and **chatbots** are **not true robots** because they have no physical movement.

Independent and dependent robots

Independent (autonomous) robot	Dependent robot
Has **no direct human control**	Has a **human interfacing directly** with it (computer or control panel)
Can **totally replace** human activity	**Supplements** (works with) human activity, does not replace it
Example: autonomous vehicle, robot vacuum cleaner	Example: car assembly line where humans and robots work together

Programming an industrial robot (for example a robot arm that paints):
1. The robot is given a **written sequence of instructions**, or
2. A **human performs the task** and the movements are **recorded**: the worker guides the arm by hand, or straps **sensors on their own arm** and sprays the object. The movements are stored as instructions, and the robot then repeats them **identically every time**.

Robots also use sensors so they do not do "stupid things", for example not spraying if **no car is present** or if the **paint has run out**.

6.2.3 Roles, Advantages and Disadvantages of Robots

Industry

Robots do heavy lifting, paint spraying, welding, making microchips and electrical goods, and work in automatic warehouses.

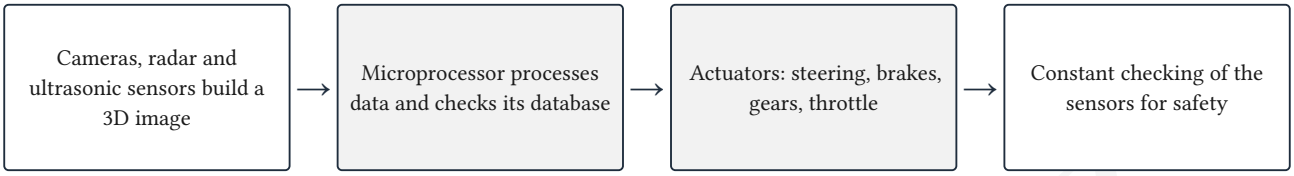
Advantages	Disadvantages
• Can work in **hazardous** conditions • Work **24/7** without a break • **Cheaper in the long run** (fewer salaries) • Higher **productivity** and very **consistent** results • Suited to **boring, repetitive** tasks	• Find **non-standard tasks** difficult (for example a cracked windscreen) • **Unemployment** in manual jobs • **Deskilling** of workers • Factories can move to cheaper countries (more unemployment)

• Less heating and lighting needed	• **Expensive** to buy and set up

> **Exam Tip:** Robots are **more consistent**, not necessarily **more accurate**. Use the word **consistent** or **repetitive** in your answer.

Transport

Autonomous cars and buses



How an autonomous car makes decisions

- **Cameras** collect visual data. **Radar and ultrasonic sensors** build a 3D image, which is very useful in fog, rain or darkness.
- **Example (red traffic light):** the car recognises the signal, checks its database, then the microprocessor tells actuators to **apply the brakes and select park**. When the light turns green the actuators **select first gear, release the brakes and operate the throttle**, but only after the sensors confirm it is safe.

Advantages	Disadvantages
• **Safer**: human error is removed • Better for the environment (efficient driving) • Less **congestion** (no stop-and-go traffic jams) • More lane capacity and shorter journey times • Stress-free parking	• Very **expensive** to set up • Could be **hacked** • Software glitches could be catastrophic • Cameras must be clean and sensors can fail in heavy snow • People are **reluctant** to trust it • Taxi drivers may lose their jobs

Autonomous trains

They use **LiDAR (light detection and ranging)**, which uses **lasers to build a 3D image** of the surroundings. They also use **proximity sensors** (on doors), **cameras** (including infrared) and **GPS** to calculate speed and direction changes. **Actuators** control speed, braking and the doors.

Advantages	Disadvantages
• Better punctuality • Lower running costs (fewer staff) • Safer: no driver error • Uses less energy • More frequent trains and easy timetable changes	• Can be hacked • Does not work well on very busy services • Passenger behaviour (jamming doors, standing near the platform edge) must be watched, so CCTV is needed • Passengers are reluctant • High starting costs

Pilotless aeroplanes

Aircraft already use autopilot. A pilotless plane would use sensors for **turbulence** and **cabin depressurisation**, **GPS** for navigation, self-testing of circuits, and actuators for the **throttle, flaps and rudder**.

Advantages	Disadvantages
• Improved passenger comfort	• Security risks (for example terrorist attacks)

• Lower running costs • Safer (most crashes are caused by pilot error) • Better aerodynamics (no cockpit needed)	• Emergencies are hard to deal with • Could be hacked • Passengers are reluctant • Software glitches can be devastating

Agriculture

Use	How robots help
Harvesting and picking	Cameras decide if a crop is ripe (for example the **Vegebot** lettuce robot). A second camera guides the arm to cut it without damage. Faster, more accurate, less waste
Weed control	Uses **AI** to tell weeds from crops. **GPS** keeps the robot on course along rows and an **actuator** operates the blade. A **drone** often maps the area first
Phenotyping	Observes physical characteristics of a plant to check **health and growth**. Uses **spectral sensors and thermal cameras** to make a 3D model, and **machine learning** to spot problems such as blight
Seed planting and fertiliser	**Drones** give an aerial image so seeds and fertiliser are placed accurately, with less waste than conventional spraying. Drones can also do **cloud seeding**
Autonomous devices	Mowers, weeding, pruning, harvesting, seeding and spraying robots use sensors and cameras to avoid obstacles and can be programmed to stop in bad weather

Advantages	Disadvantages
• Faster and more accurate than people • Replace slow, repetitive, dull work • Higher **yields** and less waste • Lower labour costs and can work 24/7 • Feed a growing world population (about nine billion by 2050)	• **Expensive** to buy and maintain • **Job losses** for farm workers • Can struggle with rough terrain and bad weather • Need skilled technicians for repairs • May be hacked or may malfunction

Medicine

- **Surgery:** safer, quicker and less costly; robots also monitor patients and do minor surgery.
- **Disinfecting** rooms and operating theatres automatically.
- **Taking blood samples:** less painful (finds a good vein), safer for staff with infectious patients, and frees staff for skilled work.
- **Microbots** (microscopic robots with a microprocessor) in **targeted therapy**: they deliver a drug to a specific site, so there is less damage to surrounding tissue.
- **Prosthetic limbs** are mini robots: they have sensors, actuators and a controller. **Neural implants and bionic skin** link to the nervous system to give better control and grip.

Disadvantages: very expensive, need specialist training and maintenance, lack of human touch, and a fault could be dangerous to a patient.

Domestic (Home) Use

- **Autonomous vacuum cleaners and lawn mowers:** proximity sensors and cameras avoid obstacles and cover the whole area, a **microprocessor** controls the device (and lets the user program it), and **actuators** drive the motors.

- **Personal assistants (for example Vector):** understand **voice commands** (microphone), answer questions using **cloud connectivity**, and use an HD camera with **computer vision** to recognise faces and avoid objects.

Advantages: saves time and effort. **Disadvantages:** expensive, sensors can miss objects, and some can be hacked.

Entertainment

- **Theme parks:** autonomous robots (often dressed as cartoon characters) interact safely with visitors.
- **Music festivals:** robots control lighting, lasers, visual effects and animation, fully synchronised with the music.
- **Film and TV:** robots control and steady **cameras**, move props and actors, and humanoid robots do **stunts** that are impossible or unsafe for people. Special effects can be timed to within a **millisecond**.

Advantages: precision and speed beyond human ability, safer than human stunts. **Disadvantages:** expensive, and jobs for stunt performers and crew may be lost.

6.3.1 What is Artificial Intelligence?

Artificial intelligence (AI) is a branch of computer science dealing with the **simulation of intelligent human behaviour by a computer**. It tries to copy the **cognitive functions** of the human brain: the mental process of **acquiring knowledge and understanding through thought, experience and the five senses**.

AI is really **a collection of rules and data** that gives a computer system the ability to **reason, learn and adapt** to external stimuli.

6.3.2 Characteristics of AI

- **Reasoning:** drawing reasoned conclusions from data and situations. In **deductive reasoning** correct facts build a set of **rules** that can be applied to new problems (for example, learning to make a perfect cup of tea and then applying the rules to coffee or hot chocolate).
- **Learning:** the system improves from experience, so next time it does the task **more effectively**.
- **Adapting:** it changes its behaviour for **new or unexpected situations**.
- It can **discern patterns** (sometimes ones humans cannot see) and **make predictions** from new data.

Three categories of AI:

Type	Meaning
Narrow AI	Performs **better than a human** at **one specific task**
General AI	**Similar** (not better) to a human when doing a specific task
Strong AI	Performs **better than a human** in **many tasks**

Examples of AI:

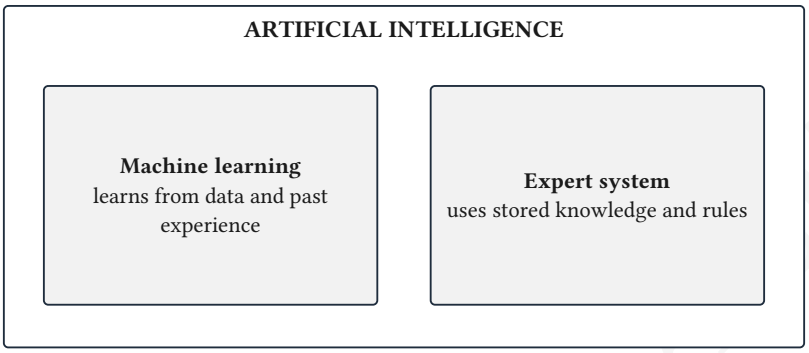
- **Smart home devices** (Alexa, Siri, Google): recognise verbal commands, convert the voice to **binary**, process it, and give a spoken answer. They **learn** from the data they receive and improve their responses.
- **Chatbots:** respond to typed or voice messages by copying human conversation, using the information known at the time.
- **Autonomous cars** (see 6.2.3).

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- **Facial expression recognition:** algorithms find key **facial landmarks** (corners of the eyebrows and mouth) and match combinations to emotions such as anger, fear, joy and surprise.
- **News generation** from live news feeds.

> **Exam Tip:** A **chatbot** is a **software** robot, so it is **not** a true robot. It is an example of AI.

6.3.3 AI Systems

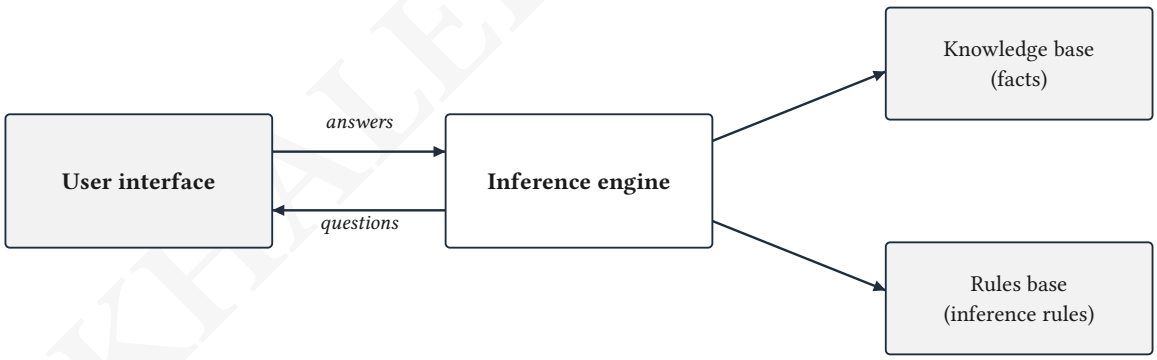


Machine learning and expert systems are both sub-sets of AI

Expert Systems

An **expert system** is an AI system that **mimics the decision-making of a human expert**. It uses **knowledge and inference** to solve problems or answer questions that normally need an expert. It gives a **conclusion**, **suggested actions** and the **percentage probability** that the conclusion is correct (for example "the probability of oil-bearing rocks at this location is about 21%").

Components of an expert system:



Structure of an expert system

Component	Purpose
User interface	How the system and user communicate (dialogue boxes, command prompts). Questions usually have **yes / no** answers and depend on the earlier answers
Inference engine	The **main processing part**. It acts like a **search engine**: it asks the user questions, searches the **knowledge base** for matching data and uses the **rules base** to reach a conclusion
Knowledge base	A **repository of facts** from experts: a collection of **objects and their attributes**

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Rules base	A set of **inference rules** (usually IF statements) used by the inference engine to draw conclusions, following human reasoning

Example knowledge base:

Object	Class	Pet?	Lives in	Body covering
Dog	Mammal	Can be	Land	Fur
Whale	Mammal	Not a pet	Water	Skin
Duck	Bird	Not a pet	Water	Feathers

If the answers are: mammal YES, pet NO, lives in water YES, covered in skin YES, has legs NO, the inference engine matches the object and concludes it is a **whale**.

Example rule: `IF continent = "South America" AND language = "Portuguese" THEN country = "Brazil"`

How an expert system gives a diagnosis (for example medical):
1. The **user interface** asks a series of questions. Each question depends on the previous answer.
2. The **inference engine** compares the answers with the **knowledge base**, using the **rules base**.
3. When it finds a match, it outputs the likely illness with a **probability**, and **suggested treatment**.
4. The user can ask for more information to narrow down the result.

Setting up an expert system:
1. Collect information from **human experts** or written sources and put it in the **knowledge base**.
2. Create the **rules base** of inference rules.
3. Set up the **inference engine**.
4. Develop the **user interface**.
5. **Test** it with known outcomes and correct any errors.

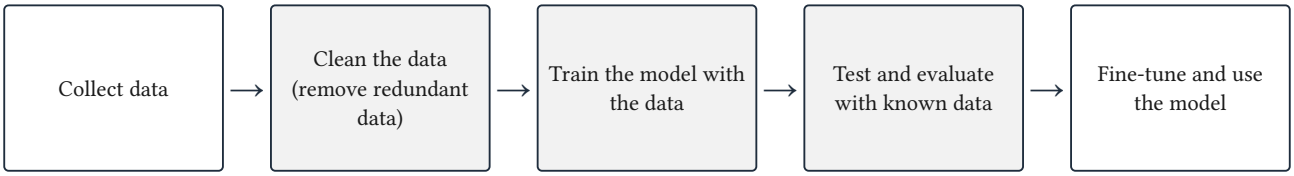
Uses: medical diagnosis, oil and mineral prospecting, fault diagnosis in machines and electronics, tax and financial calculations, strategy games such as chess, route planning for deliveries, and identifying plants, animals or chemicals.

Advantages	Disadvantages
• High level of **expertise** and accuracy • **Consistent** results • Can store huge amounts of facts • **Very fast** responses • Gives the **probability** and a traceable reason for its answer • **Unbiased** (not affected by emotion)	• Users need **training** • High set-up and maintenance costs • Responses are 'cold' (not suitable for sensitive medical news) • Only as good as the facts entered • People wrongly assume it is never wrong

Machine Learning

Machine learning is a **sub-set of AI** in which algorithms are **trained with sample data** and learn from **past experience and examples**. They can then make **predictions or decisions on new, unseen data without being specifically programmed** for that data. It can analyse huge amounts of complex data far faster than humans.

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Developing a machine learning model

Examples:

- **Spam email filter:** the algorithm collects data about emails (content, headers, sender). It **removes stop words** (the, and, a) and punctuation. Words such as "lottery" and "earn" are common in spam. The model is **trained** with past emails known to be spam, then evaluated, fine-tuned and tested live.
- **Recommendations from buying history** (collaborative filtering): customers with **similar shopping behaviour** are compared. If customer A bought a jazz CD, a Roman history book and health food, and customer B bought the first two, the system recommends the **health food** to B.
- **Detecting fraud:** data about customers' normal transactions is collected and cleaned. The model is trained with real purchase patterns and tested with known outcomes. An **unusual purchase** (for example a very expensive piece of jewellery) is flagged as possible fraud.
- **Search engines:** if the user finds the result on **page 1** it is a **success**, if the user must go to later pages it is a **failure**. The engine learns from this, so results improve over time.

Artificial intelligence	Machine learning
Simulated intelligence in machines	Getting machines to **make decisions without being programmed** to do so
Aim: build machines that **think like humans**	Aim: machines that **learn from data** so they can solve **new problems**
The **whole field**	A **sub-set** of AI

Model Answers

Describe how sensors, a microprocessor and actuators are used in an automated system (for example a greenhouse).

- Sensors (for example temperature and humidity) take readings and send the data to the microprocessor. An ADC converts analogue data to digital.
- The microprocessor compares the data with the stored values.
- If a value is outside the limit, it sends a signal to an actuator, which opens a window, turns on a heater or switches on a pump.
- The sensors keep reading, so the process is continuous.

Give three advantages and two disadvantages of using an automated system.

- Advantages: faster response than a human, safer for workers, cheaper in the long run, more consistent.
- Disadvantages: expensive to set up, can be hacked, needs expensive maintenance.

Give three characteristics of a robot.

- It senses its surroundings using sensors.
- It has a degree of movement.
- It is programmable (it has a controller).

Explain the difference between an independent and a dependent robot.

- An independent (autonomous) robot has no direct human control and can replace the human activity completely.

- A dependent robot has a human interfacing with it and supplements the human activity rather than replacing it.

Describe how an autonomous car stops at a red traffic light.
- Cameras recognise the red light and the microprocessor checks its database for the action.
- Radar and ultrasonic sensors check the surroundings.
- The microprocessor sends signals to actuators to apply the brakes and select park.
- It keeps checking until the light turns green, then releases the brakes and uses the throttle when the sensors show it is safe.

Describe the main parts of an expert system.
- The user interface lets the user answer questions and see the result.
- The inference engine searches the knowledge base for matches.
- The knowledge base is a repository of facts (objects and attributes).
- The rules base is a set of inference rules used to reach a conclusion.

Explain the difference between AI and machine learning.
- AI is the simulation of intelligent human behaviour by a computer, so that machines can think like humans.
- Machine learning is a sub-set of AI in which algorithms learn from data and past examples to make decisions on new data without being programmed for it.

Give two reasons why camera images from the train doors are still sent to the driver in the cab.
- The driver can see if passengers are trapped or in danger at the doors.
- The driver can take extra action if the computer system fails or does not react correctly.

Common Exam Questions

- Describe how sensors, microprocessor and actuators are used in a named automated system (greenhouse, irrigation, car park barrier, train doors, lighting, weather station).
- Give advantages and disadvantages of using an automated system in a given scenario.
- Name suitable sensors for a given application and say what each measures.
- Describe how a self-parking car or adaptive cruise control works.
- State what is meant by robotics and give the three characteristics of a robot.
- Explain the difference between independent and dependent robots.
- State what an end-effector is and give an example.
- Give advantages and disadvantages of robots in industry, agriculture, transport or medicine.
- Explain why a chatbot or a search bot is not a true robot.
- State what is meant by AI and name the three categories.
- Name and describe the components of an expert system.
- Describe how an expert system is used to reach a diagnosis.
- Explain what machine learning is and how it differs from AI.
- Describe how machine learning is used (spam filter, fraud detection, recommendations).

Quick Revision

> - **Automated system** = software and hardware (sensors, microprocessor, actuators) working with no human intervention. Humans may still monitor it.
> - **Cycle:** sensor reads → ADC (if analogue) → microprocessor compares with stored value → actuator acts → repeat.

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- **Advantages:** faster, safer, consistent, cheaper long term, 24/7. **Disadvantages:** expensive set-up and maintenance, hacking, untested situations, job losses.
- **DCS** = computer that controls a complex process (nuclear power station). A **supervisor** can override.
- **Self-parking and cruise control:** signals bounce back, the **time taken** gives the distance, actuators control steering, brakes and throttle.
- **Weather station** sensors: thermometer, anemometer, hygrometer, barometer, level sensor, light sensor.
- **Accelerometer** = acceleration and tilt. **Proximity sensor** = hand or finger position.
- **Robotics** = design, construction and operation of robots. **Robot** = senses, moves, programmable.
- **End-effector** = tool on a robot arm. **Independent** = no human control. **Dependent** = works with a human.
- **Search bots and chatbots** are software, so **not true robots**. Many robots have no AI.
- **Autonomous vehicles:** cameras, radar, ultrasonics, LiDAR (lasers, 3D image), GPS, microprocessor, actuators.
- **AI** = collection of rules and data to reason, learn and adapt. Types: **narrow, general, strong**.
- **Expert system** = user interface + inference engine + knowledge base + rules base. Gives answer and probability.
- **Machine learning** = sub-set of AI that learns from training data to predict on new data without being programmed for it.

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